Extremally projective normed spaces

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Abstract: A normed space is called extremally projective if it is C-topologically projective in the category of normed spaces for every $C > 1$. Helemskii proved that the spaces $\ell_1^0(\Lambda)$ of finitely supported functions on a set Λ with the ℓ_1-norm are extremally projective and asked whether every extremally projective normed space is isometrically isomorphic to some $\ell_1^0(\Lambda)$; the same conjecture was raised by Grønbæk. We answer this question in the negative and prove that the conjecture holds in the almost isometric form: a normed space P is extremally projective if and only if it is topologically projective and its completion is isometrically isomorphic to $\ell_1(\Gamma)$ for some set Γ, if and only if the Banach–Mazur distance between P and $\ell_1^0(\Gamma)$ equals 1. For spaces of countable dimension this means that P is extremally projective if and only if its completion is isometrically isomorphic to ℓ_1 or ℓ_1^n. The proof is a transfinite back-and-forth construction. Along the way we show that extremally projective normed spaces contain no infinite dimensional complete subspaces, so that no infinite dimensional Banach space is extremally projective in the category of normed spaces, and we use Fodor's lemma to construct a dense subspace of $\ell_1(\omega_1)$ all of whose separable subspaces have countable dimension but which is not topologically projective.

Keywords: projective normed space, topological projectivity, extremal projectivity, Banach–Mazur distance, ℓ_1-space.

1 Introduction

Projectivity in categories of normed and Banach modules admits several non-equivalent versions. Besides the classical relative projectivity of Helemskii [1], which ignores the norms of lifting operators, there are so-called metric and topological projectivity [2, 4, 5], where the norm of a lifting is controlled by a constant. In the metric theory this constant is 1, and in the topological theory it is some fixed $C \geq 1$. The present paper is devoted to the limiting case between these two theories: a module is called extremally projective if it is C-topologically projective for every $C > 1$. This notion was introduced and studied by Helemskii under the name of extreme projectivity [3, 2]. We restrict ourselves to the simplest possible algebra, namely the zero algebra. Normed modules over the zero algebra are just normed spaces, so we deal with extremal projectivity in the category **Nor** of normed spaces and bounded linear operators.

For Banach spaces the corresponding questions are classical. By a theorem of Köthe [6] topologically projective Banach spaces are exactly the Banach spaces topologically isomorphic to $\ell_1(\Lambda)$ for some set Λ, and by a theorem of Grothendieck [7] a Banach space has the metric lifting property if and only if it is isometrically isomorphic to some $\ell_1(\Lambda)$. For normed spaces the answers are different, because the category **Nor** has its own free objects. These are the spaces $\ell_1^0(\Lambda)$ of finitely supported functions on a set Λ with the ℓ_1-norm. Grønbæk [9] extended Köthe's theorem to **Nor**: a normed space E is topologically projective if and only if it is topologically isomorphic to $\ell_1^0(\Lambda)$, where Λ is a set whose cardinality is the density character of E. Helemskii [2] proved the metric counterpart: a normed space is metrically projective if and only if it is isometrically isomorphic to some $\ell_1^0(\Lambda)$. For extremal projectivity the following facts were known. The spaces $\ell_1^0(\Lambda)$ are extremally projective, and a normed space is extremally projective if and only if it is a near-retract of some $\ell_1^0(\Lambda)$ [3, propositions 2.2 and 2.3]. Every extremally projective normed space is, up to

an isometric isomorphism, a dense subspace of $\ell_1(\Lambda)$ for some set Λ [3, proposition 2.3], see also [9, section 3], and every separable topologically projective normed space has at most countable algebraic dimension [3, proposition 2.4]. Both Helemskii [3, section 2] and Grønbæk [9, conjecture 3.1] asked whether, conversely, every extremally projective normed space is isometrically isomorphic to some $\ell_1^0(\Lambda)$; in a more detailed form, Helemskii asked which dense subspaces of $\ell_1(\Lambda)$ are extremally projective.

The main results of this paper are the following. First, the answer to the question of Helemskii and Grønbæk is negative: the subspace of ℓ_1 spanned by the finitely supported sequences and a single sequence with infinite support is extremally projective but not isometrically isomorphic to any $\ell_1^0(\Lambda)$. Second, extremal projectivity is an almost isometric property: a normed space P is extremally projective if and only if it is topologically projective and its completion is isometrically isomorphic to $\ell_1(\Gamma)$ for some set Γ, if and only if for every $\varepsilon > 0$ there is a linear bijection $T : P \to \ell_1^0(\Gamma)$ with $\|T\|\|T^{-1}\| < 1 + \varepsilon$. Thus, the conjecture of Grønbæk holds in the almost isometric form. For spaces of countable algebraic dimension the condition of topological projectivity is superfluous: such a space is extremally projective if and only if its completion is isometrically isomorphic to ℓ_1 or to ℓ_1^n, so every dense subspace of ℓ_1 of countable algebraic dimension is extremally projective. Third, we extend the dimension restriction of Helemskii to non-separable spaces: every separable subspace of a topologically projective normed space has countable algebraic dimension, and therefore topologically projective normed spaces contain no infinite dimensional subspaces which are complete with respect to a norm dominating the original one. In particular, an infinite dimensional Banach space is never extremally projective in **Nor**, although it may well be extremally projective in the category **Ban** of Banach spaces. The last fact explains the main technical difficulty of the subject: when the lifting problems are posed for all normed spaces rather than Banach spaces, the standard tools of Banach space geometry, which rely on completeness, are no longer available, and we must work with the finite dimensional pieces of our spaces directly. Our main tool is a back-and-forth construction, which in the case of countable dimension replaces the decomposition method of Pełczyński in the proofs of Köthe and Grønbæk, and in the general case takes the form of a transfinite recursion along a filtration of Γ whose perturbations are controlled on disjoint blocks of coordinates. Fourth, in the non-separable case the dimension restriction admits a set-theoretic refinement: by Fodor's lemma, the closed linear spans of the initial segments of an ω_1-sequence of vectors in a topologically projective normed space must behave continuously at almost all limit stages. Using this we construct a dense subspace of $\ell_1(\omega_1)$, built from a ladder system on ω_1, which is not topologically projective, although all its separable subspaces have countable dimension and all its countably generated pieces are extremally projective. Thus, for uncountable Λ the question of Helemskii on dense subspaces of $\ell_1(\Lambda)$ does not reduce to the countable case: a dense subspace of $\ell_1(\Lambda)$ is extremally projective if and only if it is topologically projective, and for uncountable Λ this is a genuine restriction.

Let us fix the notation and recall the definitions. All normed spaces are considered over the complex field; the real case is identical. For a normed space E we denote by B_E its closed unit ball and by B_E° its open unit ball. By **Nor** we denote the category of normed spaces and bounded linear operators and by **Ban** its full subcategory of Banach spaces. For a set Λ we denote by $\ell_1(\Lambda)$ the Banach space of absolutely summable families and by $\ell_1^0(\Lambda)$ its dense subspace consisting of finitely supported families. For $\lambda \in \Lambda$ the symbol e_λ denotes the family which equals 1 at λ and 0 elsewhere; the family $(e_\lambda)_{\lambda\in\Lambda}$ is a Hamel basis of $\ell_1^0(\Lambda)$. For a normed space E and a set Λ there is a bijection between bounded families $(x_\lambda)_{\lambda\in\Lambda}$ in E and bounded linear operators $T : \ell_1^0(\Lambda) \to E$, given by $T(e_\lambda) = x_\lambda$, and $\|T\| = \sup_{\lambda\in\Lambda}\|x_\lambda\|$ [3, proposition 2.1], [9, proposition 1.10]. This is the sense in which the spaces $\ell_1^0(\Lambda)$ are free objects of **Nor**; note that no completeness of E is required

here, in contrast with the case of $\ell_1(\Lambda)$.

Following [5], we say that a bounded linear operator $\xi : X \to Y$ between normed spaces is c-topologically surjective, where $c \geq 1$, if $c\,\xi(B_X^\circ) \supset B_Y^\circ$, and that ξ is coisometric if it is contractive and 1-topologically surjective, that is, if $\xi(B_X^\circ) = B_Y^\circ$. Equivalently, ξ is c-topologically surjective if for every $y \in Y$ and every $\delta > 0$ there exists $x \in X$ with $\xi(x) = y$ and $\|x\| < c\|y\| + \delta$. A linear bijection $T : E \to F$ is called a c-isomorphism if $\|T\|\|T^{-1}\| \leq c$. The Banach–Mazur distance between E and F is

$$d_{BM}(E, F) = \inf\{\|T\|\|T^{-1}\| : T : E \to F \text{ is a linear bijection}\},$$

with $d_{BM}(E, F) = +\infty$ if there are no bounded linear bijections with bounded inverses between E and F. A bounded linear operator $\zeta : E \to F$ is called a c-retraction if there exists a bounded linear operator $\eta : F \to E$ with $\zeta\eta = 1_F$ and $\|\zeta\|\|\eta\| \leq c$. In this case F is called a c-retract of E.

Definition 1.1. *Let $C \geq 1$. A normed space P is called C-topologically projective if for every $c \geq 1$, every c-topologically surjective operator $\xi : X \to Y$ between normed spaces and every bounded linear operator $\phi : P \to Y$ there exists a bounded linear operator $\psi : P \to X$ such that $\xi\psi = \phi$ and $\|\psi\| \leq cC\|\phi\|$. A normed space is called topologically projective if it is C-topologically projective for some $C \geq 1$, and extremally projective if it is C-topologically projective for every $C > 1$.*

The operator ψ in definition 1.1 is called a lifting of ϕ along ξ. Extremal projectivity is precisely the extreme projectivity of Helemskii, see proposition 2.4 below. We emphasize that the spaces X and Y in definition 1.1 are arbitrary normed spaces. If we restrict them to Banach spaces, we obtain the notions of topological and extremal projectivity in **Ban**, which we shall mention only for comparison. The two notions differ substantially, see section 3.

The paper is organized as follows. In section 2 we collect the basic properties of extremally projective normed spaces: the spaces $\ell_1^0(\Lambda)$ are extremally projective, extremal projectivity passes to $(1+\varepsilon)$-retracts, and a normed space is extremally projective if and only if it is a $(1+\varepsilon)$-retract of a free normed space for every $\varepsilon > 0$. In section 3 we discuss the dimension restrictions on topologically projective normed spaces. Section 4 is devoted to the almost isometric characterization of extremally projective normed spaces of countable dimension, and section 5 contains the counterexample to the question of Helemskii and Grønbæk. Section 6 treats spaces of uncountable dimension; it contains the ladder space example and the main theorem.

2 Basic properties of extremally projective normed spaces

We start with the free objects of **Nor**.

Proposition 2.1. *Let Λ be a set. Then $\ell_1^0(\Lambda)$ is $(1+\varepsilon)$-topologically projective for every $\varepsilon > 0$. In particular, $\ell_1^0(\Lambda)$ is extremally projective.*

Proof. By [3, proposition 2.2] the space $\ell_1^0(\Lambda)$ is extremely projective in the sense of Helemskii, see proposition 2.4 below, and the proof of that proposition shows that extreme projectivity gives exactly $(1+\varepsilon)$-topological projectivity for every $\varepsilon > 0$. □

The constant $1+\varepsilon$ in proposition 2.1 cannot be replaced by 1. In fact, with the convention of definition 1.1, 1-topological projectivity is an empty notion in **Nor**.

Proposition 2.2. *The only* 1*-topologically projective normed space is the zero space.*

Proof. Let X be the space of finitely supported sequences with the sup-norm and $f : X \to \mathbb{C}$ be the functional $f(x) = s^{-1} \sum_{n=1}^{\infty} 2^{-n}(1 - 2^{-n})x_n$, where $s = \sum_{n=1}^{\infty} 2^{-n}(1 - 2^{-n})$. Clearly, $\|f\| \leq 1$. The operator f is coisometric: for $z \in B_{\mathbb{C}}^{\circ}$ choose N with $|z| < s^{-1} \sum_{n=1}^{N} 2^{-n}(1 - 2^{-n})$, then $z = f(x)$ for the sequence $x = t \operatorname{sgn}(z) \sum_{n=1}^{N} e_n$ with a suitable $t \in [0, 1)$, and $\|x\| < 1$. On the other hand, $|f(x)| = 1$ for $x \in B_X$ would force $x_n = \operatorname{sgn}(f(x))$ for all n, which is impossible for a finitely supported sequence. Thus, $f(B_X) = f(B_X^{\circ}) = B_{\mathbb{C}}^{\circ}$.

Now let P be a non-zero 1-topologically projective normed space. Take $p \in P$ with $\|p\| = 1$ and, by the Hahn–Banach theorem, a functional $g \in P^*$ with $\|g\| = g(p) = 1$. Lifting g along the coisometric operator f we get an operator $\psi : P \to X$ with $f\psi = g$ and $\|\psi\| \leq 1$. Then $x = \psi(p) \in B_X$ and $f(x) = g(p) = 1$, which contradicts the previous paragraph. □

Proposition 2.2 shows why in **Nor** it is natural to consider all constants $C > 1$ at once, instead of the single constant $C = 1$ of the metric theory.

Remark 2.3. *The proof of proposition 2.2 uses only the fact that the coisometric functional f does not attain its norm. Functionals of this kind exist on Banach spaces too, for example $x \mapsto \sum_{n=1}^{\infty}(1 - 2^{-n})x_n$ on ℓ_1. Therefore, if c-topologically surjective operators are defined via open balls as in definition 1.1, then* 1*-topological projectivity is an empty notion in* **Ban** *as well, and the free Banach spaces $\ell_1(\Lambda)$ are only $(1 + \varepsilon)$-topologically projective for every $\varepsilon > 0$. The metric theory of [2] avoids this problem by posing lifting problems along strictly coisometric operators, that is, operators mapping the closed unit ball onto the closed unit ball; with this convention $\ell_1(\Lambda)$ and $\ell_1^0(\Lambda)$ are* 1*-topologically projective in* **Ban** *and* **Nor** *respectively, and* 1*-topological projectivity coincides with metric projectivity, as in [5, proposition 2.1.9]. For the purposes of this paper the choice of convention is immaterial: if we call $\xi : X \to Y$ strictly c-topologically surjective whenever $c\xi(B_X) \supset B_Y$ and define strict C-topological projectivity by lifting along such operators with the same bound $\|\psi\| \leq cC\|\phi\|$, then every C-topologically projective space is strictly C-topologically projective, and every strictly C-topologically projective space is $C(1 + \varepsilon)$-topologically projective for every $\varepsilon > 0$. The first claim is obvious, since strictly c-topologically surjective operators are c-topologically surjective. For the second one, given a c-topologically surjective operator $\xi : X \to Y$ and $\varepsilon > 0$, endow X with the equivalent norm $\|x\|'' = \max(\|x\|, (1 + \varepsilon)c\|\xi(x)\|)$; then $(1 + \varepsilon)c\xi$ becomes a contractive operator mapping the closed unit ball onto the closed unit ball, and a lifting ψ'' along it with $\|\psi''\|'' \leq C\|\phi\|$ gives the lifting $\psi = (1 + \varepsilon)c\psi''$ along ξ with $\|\psi\| \leq (1 + \varepsilon)cC\|\phi\|$. In particular, the two conventions lead to the same notion of extremal projectivity.*

Our definition of extremal projectivity is equivalent to Helemskii's definition of extreme projectivity [3, definition I]: a normed space P is extremely projective if for every coisometric operator $\xi : X \to Y$, every bounded linear operator $\phi : P \to Y$ and every $\delta > 0$ there exists a lifting $\psi : P \to X$ of ϕ along ξ with $\|\psi\| < \|\phi\| + \delta$.

Proposition 2.4. *A normed space is extremally projective if and only if it is extremely projective in the sense of Helemskii.*

Proof. Let P be extremally projective, $\xi : X \to Y$ be a coisometric operator, $\phi : P \to Y$ be a bounded linear operator and $\delta > 0$. If $\phi = 0$ we take $\psi = 0$. Otherwise we put $C = 1 + \delta\|\phi\|^{-1}/2$. Since ξ is 1-topologically surjective, C-topological projectivity of P gives a lifting ψ of ϕ along ξ with $\|\psi\| \leq C\|\phi\| < \|\phi\| + \delta$.

Conversely, let P be extremely projective, $C > 1$, $\xi : X \to Y$ be a c-topologically surjective operator and $\phi : P \to Y$ be a non-zero bounded linear operator. Consider the operator $\xi' = c\xi$ and endow the vector space X with the norm $\|x\|' = \max(\|x\|, \|\xi'(x)\|)$, which is equivalent to the original one since $\|x\| \le \|x\|' \le \max(1, c\|\xi\|)\|x\|$. Denote the resulting normed space by X'. The operator $\xi' : X' \to Y$ is contractive by construction. If $y \in B_Y^\circ$, then by c-topological surjectivity of ξ there is $x \in B_X^\circ$ with $\xi'(x) = y$, and $\|x\|' = \max(\|x\|, \|y\|) < 1$. Thus, $\xi' : X' \to Y$ is coisometric. By extreme projectivity there is a lifting $\psi' : P \to X'$ of ϕ along ξ' with $\|\psi'\| < \|\phi\| + (C-1)\|\phi\| = C\|\phi\|$. Regarded as an operator into X, the operator $\psi = c\psi'$ satisfies $\xi\psi = \xi'\psi' = \phi$ and $\|\psi\| \le c\|\psi'\|' \le cC\|\phi\|$. Since $C > 1$ is arbitrary, P is extremally projective. □

Next we record the standard permanence property of projectivity. The proof is the same as in the Banach case, see [4, lemma 2.7] or [5, proposition 2.1.3]; for the second part see also [3, proposition 1.1].

Proposition 2.5. *Let P' be a C-topologically projective normed space and P be a c-retract of P'. Then P is cC-topologically projective. Consequently, a normed space which is a $(1+\varepsilon)$-retract of an extremally projective normed space for every $\varepsilon > 0$ is extremally projective.*

For a normed space P consider the operator

$$\pi_P : \ell_1^0(B_P) \to P, \quad e_x \mapsto x.$$

It is well defined and contractive, because the family $(x)_{x \in B_P}$ is bounded by 1. Moreover, π_P maps the closed unit ball of $\ell_1^0(B_P)$ onto B_P, since $\pi_P(e_x) = x$ for $x \in B_P$. Thus, π_P is strictly coisometric and, a fortiori, coisometric.

Proposition 2.6. *Let P be a normed space and $C \ge 1$. Then*

(i) if P is C-topologically projective, then π_P has a right inverse bounded linear operator of norm at most C;

(ii) if π_P has a right inverse bounded linear operator of norm at most C, then P is $C(1+\varepsilon)$-topologically projective for every $\varepsilon > 0$.

Proof. *(i)* Lift 1_P along the 1-topologically surjective operator π_P: we get $\sigma : P \to \ell_1^0(B_P)$ with $\pi_P\sigma = 1_P$ and $\|\sigma\| \le C$.

(ii) If $\pi_P\sigma = 1_P$ with $\|\sigma\| \le C$, then P is a C-retract of $\ell_1^0(B_P)$, which is $(1+\varepsilon)$-topologically projective by proposition 2.1. It remains to apply proposition 2.5. □

The equivalence of conditions *(i)* and *(iii)* in the following theorem is the main observation of [3, section 2] on extremally projective normed spaces: a normed space is extremally projective if and only if it is a near-retract of a free normed space, where a near-retract is a retract via a contractive operator which has right inverses of norm $< 1 + \varepsilon$ for every $\varepsilon > 0$.

Theorem 2.7. *For a normed space P the following conditions are equivalent:*

(i) P is extremally projective;

(ii) for every $\varepsilon > 0$ the operator π_P has a right inverse bounded linear operator of norm at most $1+\varepsilon$;

(iii) *for every $\varepsilon > 0$ there is a set Λ such that P is a $(1+\varepsilon)$-retract of $\ell_1^0(\Lambda)$;*

(iv) *for every $\varepsilon > 0$ there are a set Λ, a subspace $Z \subset \ell_1^0(\Lambda)$ and a projection $Q : \ell_1^0(\Lambda) \to \ell_1^0(\Lambda)$ onto Z with $\|Q\| \leq 1+\varepsilon$ such that $d_{BM}(P, Z) \leq 1+\varepsilon$.*

Proof. $(i) \implies (ii)$ is proposition 2.6 (i), and $(ii) \implies (iii)$ is obvious.

$(iii) \implies (iv)$ Let $\zeta : \ell_1^0(\Lambda) \to P$ and $\eta : P \to \ell_1^0(\Lambda)$ satisfy $\zeta\eta = 1_P$ and $\|\zeta\|\|\eta\| \leq 1+\varepsilon$. Replacing ζ by $\|\zeta\|^{-1}\zeta$ and η by $\|\zeta\|\eta$ we may assume that $\|\zeta\| = 1$ and $\|\eta\| \leq 1+\varepsilon$. Put $Z = \eta(P)$ and $Q = \eta\zeta$. Then $Q^2 = \eta(\zeta\eta)\zeta = Q$, $\mathrm{Im}(Q) = Z$ and $\|Q\| \leq 1+\varepsilon$. The operator $\eta : P \to Z$ is a bijection with $\|\eta\| \leq 1+\varepsilon$ and $\eta^{-1} = \zeta|_Z$, so $\|\eta^{-1}\| \leq 1$ and $d_{BM}(P, Z) \leq 1+\varepsilon$.

$(iv) \implies (i)$ Let $T : P \to Z$ be a linear bijection with $\|T\|\|T^{-1}\| \leq 1+\varepsilon$. Put $\eta = T$ and $\zeta = T^{-1}Q : \ell_1^0(\Lambda) \to P$. Then $\zeta\eta = T^{-1}QT = T^{-1}T = 1_P$ and $\|\zeta\|\|\eta\| \leq (1+\varepsilon)^2$. By propositions 2.1 and 2.5 the space P is $(1+\varepsilon)^3$-topologically projective. As $\varepsilon > 0$ is arbitrary, P is extremally projective. □

3 Dimension restrictions on topologically projective normed spaces

The free normed spaces $\ell_1^0(\Lambda)$ are far from being complete: for an infinite set Λ the Hamel basis $(e_\lambda)_{\lambda\in\Lambda}$ of $\ell_1^0(\Lambda)$ has the same cardinality as a dense subset of $\ell_1^0(\Lambda)$ of minimal cardinality, whereas the algebraic dimension of an infinite dimensional Banach space is uncountable by the Baire category theorem, and in fact at least continuum [10]. In this section we show that this smallness is inherited by all topologically projective normed spaces. Helemskii proved that a separable topologically projective normed space has at most countable algebraic dimension [3, proposition 2.4], and deduced that $\ell_1(\Lambda)$ is not topologically projective in **Nor** for any infinite set Λ [3, proposition 2.5]. His argument applies verbatim to subspaces of arbitrary density character, and in this hereditary form it will be convenient for us.

Theorem 3.1. *Let P be a topologically projective normed space and Z be a subspace of P. Then the algebraic dimension of Z is at most $\max(\mathrm{dens}(Z), \aleph_0)$, where $\mathrm{dens}(Z)$ is the density character of Z, that is, the smallest cardinality of a dense subset of Z. In particular, every separable subspace of P has at most countable algebraic dimension.*

Proof. By proposition 2.6 (i) there exists a bounded linear operator $\sigma : P \to \ell_1^0(B_P)$ with $\pi_P\sigma = 1_P$; in particular, σ is injective. Let M be a dense subset of Z of cardinality $\mathrm{dens}(Z)$ and $S_0 = \bigcup_{x\in M} \mathrm{supp}(\sigma(x))$; since the supports are finite, S_0 has cardinality at most $\max(\mathrm{dens}(Z), \aleph_0)$. If $z \in Z$, then $\sigma(z)$ is the limit in $\ell_1(B_P)$ of a sequence of elements of $\sigma(M)$, which are supported in S_0; since the coordinate functionals are continuous, $\sigma(z)$ is supported in S_0 as well. Thus, $\sigma(Z) \subset \ell_1^0(S_0) = \mathrm{span}\{e_s : s \in S_0\}$, a space whose algebraic dimension is the cardinality of S_0, and $\sigma|_Z$ is injective. □

Corollary 3.2. *Let P be a topologically projective normed space, Z be a subspace of P and $\|\cdot\|'$ be a norm on Z such that $(Z, \|\cdot\|')$ is complete and $\|z\| \leq K\|z\|'$ for all $z \in Z$ and some constant $K > 0$. Then Z is finite dimensional. In particular, every complete subspace of P is finite dimensional, and no infinite dimensional Banach space is topologically projective in* **Nor** *(for the spaces $\ell_1(\Lambda)$ this is [3, proposition 2.5]).*

Proof. Assume that Z is infinite dimensional. Choose a linearly independent sequence $(z_n)_{n\in\mathbb{N}}$ in Z and let Z_0 be the closure of its linear span in $(Z, \|\cdot\|')$. Then $(Z_0, \|\cdot\|')$ is a separable infinite dimensional Banach space, so its algebraic dimension is uncountable. On the other hand, Z_0 is separable with respect to $\|\cdot\|$ because $\|\cdot\| \le K\|\cdot\|'$, so by theorem 3.1 the algebraic dimension of Z_0 is countable. This contradiction proves the first claim; the rest follows by taking $Z = P$ and $\|\cdot\|' = \|\cdot\|$. □

Let us compare theorem 3.1 with the theorem of Grønbæk [9, theorem 2.1]: a normed space E is topologically projective if and only if it is topologically isomorphic to $\ell_1^0(\Lambda)$, where the cardinality of Λ is the density character of E. Since for infinite Λ the algebraic dimension of $\ell_1^0(\Lambda)$ equals the cardinality of Λ, the algebraic dimension of an infinite dimensional topologically projective normed space coincides with its density character. Theorem 3.1 is a hereditary version of this equality: it applies to all subspaces of P, which are, of course, not topologically projective in general.

The following example shows that corollary 3.2 is sharper than its last claim: a normed space may have no infinite dimensional complete subspaces and still fail to be topologically projective. It also shows that topological projectivity of a normed space is not determined by its completion.

Example 3.3. *Let $(w_n)_{n\in\mathbb{N}}$ be a sequence of real numbers with $w_n \ge 1$ and $\lim_n w_n = +\infty$, and let*

$$P_w = \Big\{ x \in \ell_1 : \sum_{n=1}^{\infty} w_n |x_n| < +\infty \Big\}$$

with the norm inherited from ℓ_1. Then P_w is a dense subspace of ℓ_1 containing $\ell_1^0(\mathbb{N})$, so its completion is ℓ_1, and

(i) every complete subspace of P_w is finite dimensional;

(ii) P_w is not topologically projective.

Proof. Denote $\|x\|_w = \sum_{n=1}^{\infty} w_n|x_n|$ for $x \in P_w$. Then $(P_w, \|\cdot\|_w)$ is the Banach space $\ell_1(w)$ of w-weighted summable sequences and $\|x\| \le \|x\|_w$ for $x \in P_w$.

(i) Let Z be a complete subspace of P_w. If $(z_k)_{k\in\mathbb{N}}$ is a sequence in Z converging to $z \in P_w$ with respect to $\|\cdot\|_w$, then it converges to z in ℓ_1, so $z \in Z$. Thus, Z is closed in $(P_w, \|\cdot\|_w)$ and the two norms $\|\cdot\|$ and $\|\cdot\|_w$ are complete on Z. Since $\|\cdot\| \le \|\cdot\|_w$, by the open mapping theorem there is a constant K with $\|z\|_w \le K\|z\|$ for all $z \in Z$. Assume that Z is infinite dimensional and fix $N \in \mathbb{N}$. The linear map $Z \to \mathbb{C}^N$, $z \mapsto (z_1, \dots, z_N)$ has a non-zero kernel, so there exists $z \in Z$ with $\|z\| = 1$ and $z_n = 0$ for $n \le N$. Then $K \ge \|z\|_w = \sum_{n>N} w_n|z_n| \ge \inf_{n>N} w_n$. Since N is arbitrary and $\lim_n w_n = +\infty$, we get a contradiction.

(ii) Apply corollary 3.2 to $Z = P_w$ and $\|\cdot\|' = \|\cdot\|_w$. Alternatively, note that P_w is separable, being dense in ℓ_1, while its algebraic dimension is uncountable, since $(P_w, \|\cdot\|_w)$ is an infinite dimensional Banach space; so theorem 3.1 applies directly. □

4 Extremally projective normed spaces of countable dimension

We start with a necessary condition which holds in all dimensions. Recall that the completion $\overline{P}$ of a normed space P is a Banach space containing P as a dense subspace. Every bounded linear operator between normed spaces extends uniquely to a bounded linear operator between

their completions with the same norm, and these extensions respect compositions. The following theorem is [3, proposition 2.3]; see also [9, section 3].

Theorem 4.1. *Let P be an extremally projective normed space. Then $\overline{P}$ is isometrically isomorphic to $\ell_1(\Gamma)$ for some set Γ.*

Proof. We only indicate the argument. By theorem 2.7 for every $\varepsilon > 0$ the space $\overline{P}$ is a $(1+\varepsilon)$-retract of the Banach space $\ell_1(B_P)$, so by Grothendieck's theorem [7] in the form given in [3, section 2] (see also [2, proposition 3.2]) $\overline{P}$ is isometrically isomorphic to $\ell_1(\Gamma)$ for some set Γ. Explicitly, taking adjoints one sees that $\overline{P}^*$ is $(1+\varepsilon)$-injective for every $\varepsilon > 0$, hence 1-injective, hence isometrically isomorphic to $C(K)$ for a Stonean space K [8, theorem 3.11.6]; a Banach space whose dual is 1-injective is isometrically isomorphic to some $L_1(\Omega, \mu)$ [8, section 17]; and $\overline{P}$ has the Schur property as a complemented subspace of $\ell_1(B_P)$, which forces μ to be purely atomic. □

Theorems 3.1 and 4.1 describe extremally projective normed spaces from two opposite sides: they are dense subspaces of the spaces $\ell_1(\Gamma)$ which are small enough to have separable subspaces of countable algebraic dimension only. In this section we show that for spaces of countable dimension there is nothing in between: every dense subspace of ℓ_1 of countable algebraic dimension is extremally projective, and even almost isometric to $\ell_1^0(\mathbb{N})$. This answers the question of Helemskii [3, section 2] on dense subspaces of $\ell_1(\Lambda)$ for countable Λ. The proof is a back-and-forth construction based on the following three lemmas. For a sequence $x \in \ell_1$ and $N \in \mathbb{N} \cup \{0\}$ we denote by $x|_N$ the sequence which coincides with x at $n \le N$ and vanishes at $n > N$.

Lemma 4.2. *Let $v \in \ell_1$, $N \in \mathbb{N} \cup \{0\}$ and $0 < \delta < 1$. Then there exists a bounded linear operator $R : \ell_1 \to \ell_1$ with $\|R\| \le \delta$ such that*

(i) $R(x) = 0$ *for every* $x \in \ell_1$ *with* $x = x|_N$;

(ii) $R(\ell_1^0(\mathbb{N})) \subset \ell_1^0(\mathbb{N})$;

(iii) $(1_{\ell_1} - R)(v) \in \ell_1^0(\mathbb{N})$;

(iv) $(1_{\ell_1} - R)^{-1}(\ell_1^0(\mathbb{N})) \subset \ell_1^0(\mathbb{N}) + \mathbb{C}v$.

In particular, $T = 1_{\ell_1} - R$ is a linear bijection of ℓ_1 with $\|T\| \le 1+\delta$ and $\|T^{-1}\| \le (1-\delta)^{-1}$ such that $T(x) = x$ whenever $x = x|_N$, $T(\ell_1^0(\mathbb{N})) \subset \ell_1^0(\mathbb{N})$ and $T(\ell_1^0(\mathbb{N}) + \mathbb{C}v) = \ell_1^0(\mathbb{N})$.

Proof. Since $\|R\| \le \delta < 1$, the operator $T = 1_{\ell_1} - R$ is a linear bijection with $T^{-1} = \sum_{m=0}^{\infty} R^m$, $\|T\| \le 1+\delta$ and $\|T^{-1}\| \le (1-\delta)^{-1}$, and the properties of T listed in the last sentence follow from (i)–(iv). So we only need to construct R with the properties (i)–(iv).

If $v - v|_N$ is finitely supported, take $R = 0$. Assume that $v - v|_N$ has infinite support. For $M \ge N$ put $t_M = \|v - v|_M\| = \sum_{n>M} |v_n|$. Then the sequence $(t_M)_{M \ge N}$ is non-increasing, $t_M > 0$ for all $M \ge N$ because $v - v|_N$ has infinite support, and $\lim_M t_M = 0$. Choose integers $N = N_0 < N_1 < N_2 < \dots$ such that $t_{N_k} \le (\delta/2) t_{N_{k-1}}$ for all $k \in \mathbb{N}$. The sets $B_k = \{n \in \mathbb{N} : N_{k-1} < n \le N_k\}$, $k \in \mathbb{N}$, are finite, pairwise disjoint and cover $\{n \in \mathbb{N} : n > N\}$. Put $w_k = v|_{N_k} - v|_{N_{k-1}}$, that is, w_k is the restriction of v to B_k. Then $w_k \in \ell_1^0(\mathbb{N})$, $v - v|_N = \sum_{k=1}^{\infty} w_k$ in ℓ_1, $\operatorname{supp}(w_k) \subset B_k$, and

$$\|w_k\| = t_{N_{k-1}} - t_{N_k} \ge (1 - \delta/2) t_{N_{k-1}} > 0, \qquad \|w_{k+1}\| \le t_{N_k} \le (\delta/2) t_{N_{k-1}} \le \frac{\delta/2}{1-\delta/2} \|w_k\| \le \delta \|w_k\|,$$

where the last inequality holds because $\delta \le 1$. For $k \in \mathbb{N}$ define the functional

$$g_k : \ell_1 \to \mathbb{C}, \quad g_k(x) = \sum_{n \in B_k} \overline{\operatorname{sgn}(v_n)}\, x_n,$$

where $\operatorname{sgn}(z) = z/|z|$ for $z \neq 0$ and $\operatorname{sgn}(0) = 0$. Then $|g_k(x)| \le \sum_{n \in B_k} |x_n|$ for all $x \in \ell_1$, $g_k(w_k) = \|w_k\|$ and $g_k(w_j) = 0$ for $j \neq k$, because the supports of w_j and g_k are disjoint. Now define

$$R : \ell_1 \to \ell_1, \quad R(x) = \sum_{k=1}^{\infty} \frac{g_k(x)}{\|w_k\|}\, w_{k+1}.$$

The series converges absolutely, since

$$\sum_{k=1}^{\infty} \frac{|g_k(x)|}{\|w_k\|} \|w_{k+1}\| \le \delta \sum_{k=1}^{\infty} \sum_{n \in B_k} |x_n| = \delta \|x\|,$$

so R is a well defined bounded linear operator with $\|R\| \le \delta$. We verify the properties (i)–(iv).

(i) If $x = x|_N$, then $g_k(x) = 0$ for all k, because $\operatorname{supp}(g_k) \subset B_k \subset \{n : n > N\}$. Hence $R(x) = 0$.

(ii) If $x \in \ell_1^0(\mathbb{N})$, then $g_k(x) \neq 0$ only for the finitely many k with $B_k \cap \operatorname{supp}(x) \neq \varnothing$, so $R(x)$ is a finite linear combination of the finitely supported sequences w_{k+1}.

(iii) Since $g_k(v|_N) = 0$ and $g_k(v - v|_N) = \sum_{j=1}^{\infty} g_k(w_j) = g_k(w_k) = \|w_k\|$, we get $R(v) = \sum_{k=1}^{\infty} w_{k+1} = v - v|_N - w_1$, so $(1_{\ell_1} - R)(v) = v|_N + w_1 \in \ell_1^0(\mathbb{N})$.

(iv) Let $x \in \ell_1^0(\mathbb{N})$ and let K be the finite set of those k for which $g_k(x) \neq 0$. Put $c_k = g_k(x)/\|w_k\|$ for $k \in K$. Since $g_j(w_k) = \delta_{jk}\|w_k\|$, we have $R(w_k) = w_{k+1}$ for all k, hence $R^m(x) = \sum_{k \in K} c_k w_{k+m}$ for $m \ge 1$ and

$$(1_{\ell_1} - R)^{-1}(x) = \sum_{m=0}^{\infty} R^m(x) = x + \sum_{k \in K} c_k \sum_{m=1}^{\infty} w_{k+m} = x + \sum_{k \in K} c_k (v - v|_{N_k}) \in \ell_1^0(\mathbb{N}) + \mathbb{C}v.$$

□

Lemma 4.3. *Let Y be a finite dimensional subspace of $\ell_1^0(\mathbb{N})$ and $n \in \mathbb{N}$ be such that $e_n \notin Y$. Then there is a constant $C \ge 1$ such that for every $z \in \ell_1^0(\mathbb{N})$ with $\|z - e_n\| \le (4C)^{-1}$ there exists a linear bijection $S : \ell_1 \to \ell_1$ with $\|S - 1_{\ell_1}\| \le 2C\|z - e_n\| \le 1/2$, $S(y) = y$ for all $y \in Y$ and $S(z) = e_n$.*

Proof. Since Y is finite dimensional, it is closed in ℓ_1, so $d = \inf\{\|e_n - y\| : y \in Y\} > 0$. By the Hahn–Banach theorem there is $f \in \ell_1^*$ with $f(y) = 0$ for $y \in Y$, $f(e_n) = 1$ and $\|f\| = d^{-1}$. Put $C = \max(1, d^{-1})$. Let $z \in \ell_1^0(\mathbb{N})$ satisfy $\|z - e_n\| \le (4C)^{-1}$. Then $|f(z) - 1| = |f(z - e_n)| \le C\|z - e_n\| \le 1/4$, so $|f(z)| \ge 3/4$. Define

$$S : \ell_1 \to \ell_1, \quad S(x) = x + \frac{f(x)}{f(z)}(e_n - z).$$

Then $S(y) = y$ for $y \in Y$ because f vanishes on Y, and $S(z) = z + (e_n - z) = e_n$. Moreover, $\|S - 1_{\ell_1}\| \le \|f\||f(z)|^{-1}\|e_n - z\| \le \frac{4}{3}C\|z - e_n\| \le 2C\|z - e_n\| \le 1/2$, so S is a linear bijection of ℓ_1 by the Neumann series. □

Lemma 4.4. *Let X be a Banach space and $(F_m)_{m\in\mathbb{N}}$ be a sequence of bounded linear operators on X such that $\theta_m = \|F_m - 1_X\| < 1$ for all m and $\sum_{m=1}^{\infty}\theta_m < +\infty$. Put $A_j = F_jF_{j-1}\cdots F_1$ for $j\in\mathbb{N}$. Then the sequence $(A_j)_{j\in\mathbb{N}}$ converges in the operator norm to a linear bijection T of X with*

$$\|T\| \le \prod_{m=1}^{\infty}(1+\theta_m), \qquad \|T^{-1}\| \le \prod_{m=1}^{\infty}(1-\theta_m)^{-1}.$$

Proof. Both products converge because $\sum_m \theta_m < +\infty$; denote them by M and M'. Then $\|A_j\| \le M$ and, since each F_m is invertible with $\|F_m^{-1}\| \le (1-\theta_m)^{-1}$ by the Neumann series, $\|A_j^{-1}\| \le M'$ for all j. From $\|A_j - A_{j-1}\| = \|(F_j - 1_X)A_{j-1}\| \le \theta_j M$ we see that $(A_j)_j$ is a Cauchy sequence; let T be its limit, so $\|T\| \le M$. Also $\|A_j^{-1} - A_k^{-1}\| = \|A_j^{-1}(A_k - A_j)A_k^{-1}\| \le M'^2\|A_k - A_j\|$, so $(A_j^{-1})_j$ is a Cauchy sequence with a limit B, $\|B\| \le M'$. Passing to the limit in $A_jA_j^{-1} = A_j^{-1}A_j = 1_X$ we get $TB = BT = 1_X$. □

Theorem 4.5. *Let D be a dense subspace of ℓ_1 of countable algebraic dimension and $\varepsilon > 0$. Then there exists a linear bijection $T : \ell_1 \to \ell_1$ with $\|T\|\|T^{-1}\| \le 1+\varepsilon$ and $T(D) = \ell_1^0(\mathbb{N})$. In particular, $d_{BM}(D, \ell_1^0(\mathbb{N})) = 1$.*

Proof. Since ℓ_1 is infinite dimensional and D is dense in ℓ_1, the algebraic dimension of D is countably infinite. Let $(d_j)_{j\in\mathbb{N}}$ be a Hamel basis of D. Choose a sequence $(\theta_m)_{m\in\mathbb{N}}$ of numbers in $(0, 1/2]$ with $\prod_{m=1}^{\infty}(1+\theta_m)(1-\theta_m)^{-1} \le 1+\varepsilon$.

We construct by induction linear bijections $F_1, F_2, \ldots$ of ℓ_1 with $\|F_m - 1_{\ell_1}\| \le \theta_m$ and finite dimensional subspaces $Y_0 = \{0\} \subset Y_1 \subset Y_2 \subset \ldots$ of $\ell_1^0(\mathbb{N})$ such that, with $A_0 = 1_{\ell_1}$ and $A_j = F_{2j}F_{2j-1}A_{j-1}$, for every $j\in\mathbb{N}$ the following conditions hold:

(a) $A_j(d_i) \in Y_j$ for $i \le j$;

(b) $e_i \in Y_j$ for $i \le j$;

(c) $Y_j \subset A_j(D)$;

(d) F_{2j-1} and F_{2j} act as the identity on Y_{j-1}.

Suppose that A_{j-1} and Y_{j-1} have been constructed (for $j = 1$ there is nothing to suppose). Let N be an integer not smaller than every element of the supports of the vectors of some basis of Y_{j-1}, so that $y = y|_N$ for all $y \in Y_{j-1}$. Apply lemma 4.2 to the vector $v = A_{j-1}(d_j)$, the integer N and $\delta = \theta_{2j-1}$, and let $F_{2j-1} = 1_{\ell_1} - R$ be the resulting operator. Then $\|F_{2j-1} - 1_{\ell_1}\| \le \theta_{2j-1}$, F_{2j-1} acts as the identity on Y_{j-1}, and $y_j = F_{2j-1}A_{j-1}(d_j) \in \ell_1^0(\mathbb{N})$. Put $A' = F_{2j-1}A_{j-1}$ and $Y' = Y_{j-1} + \mathbb{C}y_j \subset \ell_1^0(\mathbb{N})$. Note that $Y' \subset A'(D)$: indeed, $Y_{j-1} \subset A_{j-1}(D)$ by the induction hypothesis (for $j = 1$ trivially), F_{2j-1} fixes Y_{j-1}, so $Y_{j-1} = F_{2j-1}(Y_{j-1}) \subset A'(D)$, and $y_j = A'(d_j) \in A'(D)$.

If $e_j \in Y'$, put $F_{2j} = 1_{\ell_1}$ and $Y_j = Y'$. Otherwise, let C be the constant given by lemma 4.3 for Y' and $n = j$, and let N' be an integer not smaller than j and than every element of the supports of the vectors of some basis of Y'. Since A' is a linear bijection of ℓ_1 and D is dense, $A'(D)$ is dense in ℓ_1, so there is $d \in D$ with

$$\|A'(d) - e_j\| \le \frac{\theta_{2j}}{16C}.$$

Apply lemma 4.2 to the vector $A'(d)$, the integer N' and $\delta = \theta_{2j}/2$, and let $G = 1_{\ell_1} - R'$ be the resulting operator. Then G fixes Y' and e_j, $\|G - 1_{\ell_1}\| \le \theta_{2j}/2$, $\|G\| \le 5/4$ because $\theta_{2j} \le 1/2$, and $z = GA'(d) \in \ell_1^0(\mathbb{N})$ satisfies

$$\|z - e_j\| = \|G(A'(d) - e_j)\| \le \frac{5\theta_{2j}}{64C} \le \frac{1}{4C}.$$

Let S be the operator given by lemma 4.3 for Y', $n = j$ and z. Then S fixes Y', $S(z) = e_j$ and $\|S - 1_{\ell_1}\| \le 2C\|z - e_j\| \le 5\theta_{2j}/32$. Put $F_{2j} = SG$ and $Y_j = Y' + \mathbb{C}e_j$. Then

$$\|F_{2j} - 1_{\ell_1}\| \le \|(S - 1_{\ell_1})G\| + \|G - 1_{\ell_1}\| \le \frac{5\theta_{2j}}{32} \cdot \frac{5}{4} + \frac{\theta_{2j}}{2} \le \theta_{2j}.$$

Let us check (a)–(d) for j. The operator F_{2j} acts as the identity on $Y' \supset Y_{j-1}$, which together with the first step gives (d). For $i < j$ we have $A_{j-1}(d_i) \in Y_{j-1}$, so $A_j(d_i) = F_{2j}F_{2j-1}A_{j-1}(d_i) = A_{j-1}(d_i) \in Y_j$, and $A_j(d_j) = F_{2j}(y_j) = y_j \in Y_j$; this is (a). Condition (b) holds by the definition of Y_j and the induction hypothesis. Finally, $Y' \subset A'(D)$ and F_{2j} fixes Y', so $Y' \subset F_{2j}A'(D) = A_j(D)$, and $e_j = SGA'(d) = A_j(d) \in A_j(D)$; this is (c).

By lemma 4.4 the operators A_j converge in norm to a linear bijection T of ℓ_1 with $\|T\|\|T^{-1}\| \le \prod_m (1+\theta_m)(1-\theta_m)^{-1} \le 1+\varepsilon$. For every $i \in \mathbb{N}$ and $j \ge i$ we have $A_j(d_i) \in Y_j$ by (a), and by (d) the operators $F_{2j+1}, F_{2j+2}, \dots$ fix Y_j, so $A_k(d_i) = A_j(d_i)$ for all $k \ge j$ and $T(d_i) = A_i(d_i) \in Y_i \subset \ell_1^0(\mathbb{N})$. Hence $T(D) \subset \ell_1^0(\mathbb{N})$. Similarly, for every $i \in \mathbb{N}$ by (b) and (c) there is $d \in D$ with $A_i(d) = e_i \in Y_i$, and by (d) we get $T(d) = A_i(d) = e_i$. Hence $\ell_1^0(\mathbb{N}) \subset T(D)$. □

Now we are ready to prove the main result of the paper.

Theorem 4.6. *Let P be a normed space of countable (finite or countably infinite) algebraic dimension. Then the following conditions are equivalent:*

(i) P is extremally projective;

(ii) $\overline{P}$ is isometrically isomorphic to $\ell_1(\Gamma)$ for some countable set Γ;

(iii) $d_{BM}(P, \ell_1^0(\Gamma)) = 1$ for some countable set Γ;

(iv) for every $\varepsilon > 0$ there is a set Λ with $d_{BM}(P, \ell_1^0(\Lambda)) < 1 + \varepsilon$.

Proof. $(i) \implies (ii)$ By theorem 4.1 the space $\overline{P}$ is isometrically isomorphic to $\ell_1(\Gamma)$ for some set Γ. The rational linear span of a countable Hamel basis of P is a countable dense subset of $\overline{P}$, so $\overline{P}$ is separable and Γ is countable.

$(ii) \implies (iii)$ Let $J : \overline{P} \to \ell_1(\Gamma)$ be an isometric isomorphism. If Γ is finite, then $\overline{P}$ is finite dimensional, so $P = \overline{P}$ and J maps P onto $\ell_1(\Gamma) = \ell_1^0(\Gamma)$. If Γ is infinite, we may assume that $\Gamma = \mathbb{N}$. Then $D = J(P)$ is a dense subspace of ℓ_1 of countable algebraic dimension, and by theorem 4.5 $d_{BM}(P, \ell_1^0(\mathbb{N})) = d_{BM}(D, \ell_1^0(\mathbb{N})) = 1$.

$(iii) \implies (iv)$ is obvious.

$(iv) \implies (i)$ Fix $\varepsilon > 0$ and a linear bijection $S : P \to \ell_1^0(\Lambda)$ with $\|S\|\|S^{-1}\| < 1 + \varepsilon$. Then P is a $(1+\varepsilon)$-retract of $\ell_1^0(\Lambda)$ via S^{-1} and S, so P is extremally projective by theorem 2.7. □

Corollary 4.7. *Every subspace of ℓ_1 of countable algebraic dimension which is dense in ℓ_1 is extremally projective and almost isometric to $\ell_1^0(\mathbb{N})$, that is, has Banach–Mazur distance 1 to $\ell_1^0(\mathbb{N})$.*

5 Extremally projective spaces which are not free

In this section we show that extremal projectivity is not an isometric invariant of the form "P is isometrically isomorphic to $\ell_1^0(\Lambda)$". By corollary 4.7, for every subspace V of ℓ_1 of countable algebraic dimension the space $\ell_1^0(\mathbb{N}) + V$ is extremally projective. To see that such spaces are not isometrically isomorphic to free normed spaces whenever $V \not\subset \ell_1^0(\mathbb{N})$, we look at extreme points of unit balls. By $\operatorname{ext}(B)$ we denote the set of extreme points of a convex set B and by $\operatorname{conv}(M)$ the convex hull of a set M.

Lemma 5.1. *Let S be a set and E be a subspace of $\ell_1(S)$ containing $\ell_1^0(S)$, endowed with the norm of $\ell_1(S)$. Then $\operatorname{ext}(B_E) = \{\lambda e_s : s \in S,\ \lambda \in \mathbb{C},\ |\lambda| = 1\}$ and $\operatorname{conv}(\operatorname{ext}(B_E)) = B_{\ell_1^0(S)}$.*

Proof. The points λe_s with $|\lambda| = 1$ are extreme points of $B_{\ell_1(S)}$, and $B_E = B_{\ell_1(S)} \cap E$, so they are extreme points of B_E. Conversely, let $x \in B_E$ be not of this form. If $x = 0$, then $x = \frac{1}{2}e_s + \frac{1}{2}(-e_s)$ for any $s \in S$. If $0 < \|x\| < 1$, then $x = \frac{1}{2}(1+t)x + \frac{1}{2}(1-t)x$ for $t = 1 - \|x\| > 0$, and both $(1 \pm t)x$ belong to B_E. If $\|x\| = 1$, then x has at least two non-zero coordinates x_n and x_m, $n \neq m$. Put $t = \min(|x_n|, |x_m|) > 0$ and $u = t(\operatorname{sgn}(x_n)e_n - \operatorname{sgn}(x_m)e_m) \in \ell_1^0(S) \subset E$. Then $x \pm u \in E$ and

$$\|x \pm u\| = (|x_n| \pm t) + (|x_m| \mp t) + \sum_{s \neq n,m} |x_s| = \|x\| = 1,$$

so $x = \frac{1}{2}(x+u) + \frac{1}{2}(x-u)$ with $u \neq 0$. In all three cases x is not an extreme point of B_E.

Every finite convex combination of the points λe_s is a finitely supported element of $B_{\ell_1(S)}$, so $\operatorname{conv}(\operatorname{ext}(B_E)) \subset B_{\ell_1^0(S)}$. Conversely, if $x \in B_{\ell_1^0(S)}$, then

$$x = \sum_{s \in \operatorname{supp}(x)} |x_s| \operatorname{sgn}(x_s) e_s + (1 - \|x\|) \cdot 0,$$

and $0 = \frac{1}{2}e_s + \frac{1}{2}(-e_s)$, so $x \in \operatorname{conv}(\operatorname{ext}(B_E))$. □

Theorem 5.2. *Let S be a set and E be a subspace of $\ell_1(S)$ containing $\ell_1^0(S)$, endowed with the norm of $\ell_1(S)$. If E is isometrically isomorphic to $\ell_1^0(\Lambda)$ for some set Λ, then $E = \ell_1^0(S)$.*

Proof. Let $T : \ell_1^0(\Lambda) \to E$ be an isometric isomorphism. Then $T(B_{\ell_1^0(\Lambda)}) = B_E$ and T maps $\operatorname{ext}(B_{\ell_1^0(\Lambda)})$ onto $\operatorname{ext}(B_E)$. Applying lemma 5.1 to E and to $\ell_1^0(\Lambda)$ itself we get

$$B_E = T(B_{\ell_1^0(\Lambda)}) = T(\operatorname{conv}(\operatorname{ext}(B_{\ell_1^0(\Lambda)}))) = \operatorname{conv}(T(\operatorname{ext}(B_{\ell_1^0(\Lambda)}))) = \operatorname{conv}(\operatorname{ext}(B_E)) = B_{\ell_1^0(S)}.$$

Hence $E = \ell_1^0(S)$. □

Corollary 5.3. *Let V be a subspace of ℓ_1 of countable algebraic dimension with $V \not\subset \ell_1^0(\mathbb{N})$, for example $V = \mathbb{C}v$ for $v = (2^{-n})_{n \in \mathbb{N}}$. Then $\ell_1^0(\mathbb{N}) + V$ is an extremally projective normed space which is not isometrically isomorphic to $\ell_1^0(\Lambda)$ for any set Λ.*

Proof. The space $\ell_1^0(\mathbb{N}) + V$ is dense in ℓ_1 and has countable algebraic dimension, so it is extremally projective by corollary 4.7, and it is not isometrically isomorphic to any $\ell_1^0(\Lambda)$ by theorem 5.2. □

Corollary 5.3 answers the question of Helemskii from [3, section 2] and [2, section 1] in the negative and disproves [9, conjecture 3.1]. Note that by theorem 5.2 every subspace E with $\ell_1^0(\mathbb{N}) \subsetneq E \subset \ell_1$ fails to be isometrically isomorphic to a free normed space, but by example 3.3 not all of these subspaces are extremally projective. Thus, between $\ell_1^0(\mathbb{N})$ and ℓ_1 there are both extremally projective and non-projective normed spaces, and the difference between them is invisible on the level of completions. The next section provides a criterion which distinguishes them.

6 Extremally projective normed spaces of uncountable dimension

Condition (ii) of theorem 4.6 shows that for normed spaces of countable dimension extremal projectivity is a property of the completion. By example 3.3 this is no longer true for spaces of uncountable dimension: the dimension restriction of theorem 3.1 is an independent necessary condition.

By theorems 3.1 and 4.1, an extremally projective normed space is, up to an isometric isomorphism, a dense subspace of some $\ell_1(\Gamma)$ all of whose separable subspaces have countable algebraic dimension; by the theorem of Grønbæk it is also topologically isomorphic to $\ell_1^0(\Gamma)$. For countable Γ the first two conditions are also sufficient by theorems 3.1 and 4.6. We shall now show that for uncountable Γ they are not. The obstruction is a set-theoretic refinement of theorem 3.1. Recall that a subset of the first uncountable ordinal ω_1 is called closed unbounded if it is unbounded in ω_1 and contains the supremum of each of its countable subsets, and stationary if it intersects every closed unbounded set. The set of limit ordinals below ω_1 is closed unbounded. We shall use Fodor's lemma [11, theorem 8.7]: if $E \subset \omega_1$ is stationary and $\gamma : E \to \omega_1$ satisfies $\gamma(\alpha) < \alpha$ for all $\alpha \in E$, then γ is constant on some stationary subset of E. For a subset M of a normed space P we denote by $\overline{\operatorname{span}}(M)$ the closure of $\operatorname{span}(M)$ in P.

Theorem 6.1. *Let P be a topologically projective normed space, $(X_\beta)_{\beta<\omega_1}$ be a family of countable subsets of P, E be a stationary set of limit ordinals below ω_1, and $(v_\alpha)_{\alpha\in E}$ be a family in P such that $v_\alpha \in \overline{\operatorname{span}}\big(\bigcup_{\beta<\alpha} X_\beta\big)$ for every $\alpha \in E$. Then there is a stationary set $E' \subset E$ such that $\operatorname{span}\{v_\alpha : \alpha \in E'\}$ has countable algebraic dimension. In particular, the family $(v_\alpha)_{\alpha\in E}$ is linearly dependent.*

Proof. By proposition 2.6 (i) there is a bounded linear operator $\sigma : P \to \ell_1^0(B_P)$ with $\pi_P\sigma = 1_P$; it is injective. For $\alpha < \omega_1$ put

$$S_\alpha = \bigcup_{\beta<\alpha} \bigcup_{x\in X_\beta} \operatorname{supp}(\sigma(x)),$$

a countable subset of B_P. Let $\alpha \in E$. Then $\sigma(v_\alpha)$ lies in the closure in $\ell_1(B_P)$ of $\sigma\big(\operatorname{span}\bigcup_{\beta<\alpha} X_\beta\big) \subset \ell_1^0(S_\alpha)$, and since the coordinate functionals are continuous, $\sigma(v_\alpha)$ is supported in S_α. Thus, the finite set $F_\alpha = \operatorname{supp}(\sigma(v_\alpha))$ is contained in $S_\alpha = \bigcup_{\beta<\alpha} S_{\beta+1}$, and since α is a limit ordinal there is $\gamma(\alpha) < \alpha$ with $F_\alpha \subset S_{\gamma(\alpha)}$. By Fodor's lemma there are a stationary set $E' \subset E$ and $\gamma_0 < \omega_1$ with $\gamma(\alpha) = \gamma_0$ for all $\alpha \in E'$. Then $\sigma(v_\alpha) \in \ell_1^0(S_{\gamma_0})$ for all $\alpha \in E'$. The space $\ell_1^0(S_{\gamma_0})$ has countable algebraic dimension and σ is injective, so $\operatorname{span}\{v_\alpha : \alpha \in E'\}$ has countable algebraic dimension. As E' is uncountable, the family $(v_\alpha)_{\alpha\in E'}$, and a fortiori $(v_\alpha)_{\alpha\in E}$, is linearly dependent. □

Corollary 6.2. *Let P be a topologically projective normed space and $(X_\beta)_{\beta<\omega_1}$ be a family of countable subsets of P. For $\alpha < \omega_1$ put $D_\alpha = \overline{\operatorname{span}}\big(\bigcup_{\beta<\alpha} X_\beta\big)$. Then the set of limit ordinals $\alpha < \omega_1$ for which $D_\alpha \neq \bigcup_{\alpha'<\alpha} D_{\alpha'}$ is not stationary.*

Proof. Assume that this set E is stationary and for every $\alpha \in E$ choose $v_\alpha \in D_\alpha \setminus \bigcup_{\alpha'<\alpha} D_{\alpha'}$. The family $(v_\alpha)_{\alpha\in E}$ is linearly independent. Indeed, let $\sum_{\alpha\in F} c_\alpha v_\alpha = 0$ for a finite set $F \subset E$ and non-zero scalars c_α, and let $\alpha_0 = \max F$. Since $v_{\alpha_0} \neq 0$, the set $F \setminus \{\alpha_0\}$ is non-empty; put $\alpha_1 = \max(F \setminus \{\alpha_0\}) < \alpha_0$. The spaces D_α increase with α, so $v_{\alpha_0} = -c_{\alpha_0}^{-1} \sum_{\alpha\in F\setminus\{\alpha_0\}} c_\alpha v_\alpha \in D_{\alpha_1}$, which contradicts the choice of v_{α_0}. Now theorem 6.1 gives a contradiction. □

Theorem 3.1 says that the closed linear span of countably many vectors of a topologically projective space has countable dimension; corollary 6.2 says that, in addition, the closed linear spans of initial segments of an ω_1-sequence of vectors must behave continuously at almost all limit stages. The following example shows that the second condition is not implied by the first one, even for dense subspaces of $\ell_1(\Gamma)$.

Example 6.3. *For every limit ordinal $\alpha < \omega_1$ fix a strictly increasing sequence $(\alpha_n)_{n\in\mathbb{N}}$ of ordinals with $\sup_n \alpha_n = \alpha$ and put $v_\alpha = \sum_{n=1}^{\infty} 2^{-n} e_{\alpha_n} \in \ell_1(\omega_1)$. Let*

$$P_L = \ell_1^0(\omega_1) + \operatorname{span}\{v_\alpha : \alpha < \omega_1 \text{ is a limit ordinal}\} \subset \ell_1(\omega_1)$$

with the norm inherited from $\ell_1(\omega_1)$. Then

(i) P_L is dense in $\ell_1(\omega_1)$;

(ii) for every countable ordinal δ the space $P_L \cap \ell_1(\delta) = \ell_1^0(\delta) + \operatorname{span}\{v_\alpha : \alpha \le \delta \text{ is a limit ordinal}\}$ is an extremally projective normed space of countable algebraic dimension; in particular, every separable subspace of P_L has countable algebraic dimension;

(iii) P_L is not topologically projective.

Proof. *(i)* is obvious, since $P_L \supset \ell_1^0(\omega_1)$.

First we note that for limit ordinals $\alpha < \beta$ the set $\operatorname{supp}(v_\alpha) \cap \operatorname{supp}(v_\beta)$ is finite: $\operatorname{supp}(v_\alpha) \subset \alpha$, while only finitely many β_n are smaller than α, because $\sup_n \beta_n = \beta > \alpha$.

(ii) Let δ be a countable ordinal and let $x = y + \sum_{\alpha\in F} c_\alpha v_\alpha \in P_L \cap \ell_1(\delta)$, where $y \in \ell_1^0(\omega_1)$, F is a finite set of limit ordinals and $c_\alpha \neq 0$ for $\alpha \in F$. Suppose that $F \not\subset [0,\delta]$ and put $\alpha_0 = \max F > \delta$. The set $\operatorname{supp}(v_{\alpha_0}) \cap [\delta, \alpha_0)$ is infinite, because $\sup_n \alpha_{0,n} = \alpha_0 > \delta$, while the sets $\operatorname{supp}(y)$ and $\operatorname{supp}(v_\alpha) \cap \operatorname{supp}(v_{\alpha_0})$, $\alpha \in F \setminus \{\alpha_0\}$, are finite. Hence x has infinitely many non-zero coordinates in $[\delta, \alpha_0)$, which contradicts $x \in \ell_1(\delta)$. Therefore $F \subset [0,\delta]$, and then also $\operatorname{supp}(y) \subset \delta$. This proves the formula for $P_L \cap \ell_1(\delta)$. The space $P_L \cap \ell_1(\delta)$ has countable algebraic dimension, and it is dense in $\ell_1(\delta)$, because it contains $\ell_1^0(\delta)$. By theorem 4.6 it is extremally projective. Finally, if Z is a separable subspace of P_L, then the supports of the elements of Z are contained in a countable subset of ω_1, hence in some countable ordinal δ, so $Z \subset P_L \cap \ell_1(\delta)$ has countable algebraic dimension.

(iii) Let E be the set of limit ordinals below ω_1; it is closed unbounded, hence stationary. For $\alpha \in E$ we have $v_\alpha = \lim_N \sum_{n=1}^{N} 2^{-n} e_{\alpha_n} \in \overline{\operatorname{span}}\{e_\beta : \beta < \alpha\}$. The family $(v_\alpha)_{\alpha\in E}$ is linearly independent: if $\sum_{\alpha\in F} c_\alpha v_\alpha = 0$ with $F \subset E$ finite and $c_\alpha \neq 0$, then for $\alpha_0 = \max F$ and all sufficiently large n the coordinate of the left hand side at $\alpha_{0,n}$ equals $2^{-n} c_{\alpha_0} \neq 0$, because the supports of the other v_α meet $\operatorname{supp}(v_{\alpha_0})$ in finite sets. By theorem 6.1, applied to $X_\beta = \{e_\beta\}$, the space P_L is not topologically projective. □

Thus, a dense subspace of $\ell_1(\Gamma)$ all of whose separable subspaces have countable algebraic dimension need not be topologically projective, and the answer to the question of Helemskii on dense subspaces of $\ell_1(\Lambda)$ is more involved for uncountable Λ. Note that the space P_L is "locally" extremally projective: all its countably generated pieces $P_L \cap \ell_1(\delta)$ are extremally projective. So extremal projectivity of a normed space is not determined by its separable subspaces.

We now turn to positive results. The obstruction in example 6.3 is the discontinuity of the chain $P_L \cap \ell_1(\delta)$ at limit stages. We shall show that for topologically projective spaces such discontinuities can always be avoided, and that this leads to a complete description of extremally projective

normed spaces of arbitrary dimension. The proof is a transfinite version of theorem 4.5 combined with an induction on the cardinality of Γ. The key point of the transfinite construction is that the perturbations used at different steps are controlled on pairwise disjoint blocks of coordinates, so that their "product" has norm bounded independently of the length of the construction.

Let Γ be a set, λ be an ordinal and $(\Gamma_\delta)_{\delta\le\lambda}$ be an increasing family of subsets of Γ such that $\Gamma_0=\varnothing$, $\Gamma_\lambda=\Gamma$ and $\Gamma_\delta=\bigcup_{\delta'<\delta}\Gamma_{\delta'}$ for every limit ordinal $\delta\le\lambda$. We call such a family a filtration of Γ and put $N_\delta=\Gamma_{\delta+1}\setminus\Gamma_\delta$ for $\delta<\lambda$. Then $\Gamma_\delta=\bigsqcup_{\delta'<\delta}N_{\delta'}$ for all $\delta\le\lambda$, and $\ell_1(\Gamma_\delta)$ is the ℓ_1-sum of the spaces $\ell_1(N_{\delta'})$, $\delta'<\delta$. For $x\in\ell_1(\Gamma)$ and $A\subset\Gamma$ we denote by $x|_A$ the restriction of x to A, extended by zero; note that $x|_A$ need not belong to a given subspace $P\ni x$. For a subspace $P\subset\ell_1(\Gamma)$ we put $P_\delta=P\cap\ell_1(\Gamma_\delta)$ and

$$D_\delta=\{x|_{N_\delta}:x\in P_{\delta+1}\}\subset\ell_1(N_\delta),\qquad\delta<\lambda.$$

Since the kernel of the restriction map $x\mapsto x|_{N_\delta}$ on $P_{\delta+1}$ is $P_{\delta+1}\cap\ell_1(\Gamma_\delta)=P_\delta$, the space D_δ is isomorphic as a vector space to $P_{\delta+1}/P_\delta$.

Theorem 6.4. *Let Γ be a set, P be a dense subspace of $\ell_1(\Gamma)$, $(\Gamma_\delta)_{\delta\le\lambda}$ be a filtration of Γ and $0<\theta<1/2$. Suppose that*

(i) P_δ is dense in $\ell_1(\Gamma_\delta)$ for every $\delta\le\lambda$;

(ii) $P_\delta=\bigcup_{\delta'<\delta}P_{\delta'}$ for every limit ordinal $\delta\le\lambda$;

(iii) for every $\delta<\lambda$ there is a linear bijection U_δ of $\ell_1(N_\delta)$ with $U_\delta(D_\delta)=\ell_1^0(N_\delta)$, $\|U_\delta\|\le1+\theta$ and $\|U_\delta^{-1}\|\le1$.

Then there exists a linear bijection $T:\ell_1(\Gamma)\to\ell_1(\Gamma)$ with $T(P)=\ell_1^0(\Gamma)$, $\|T\|\le(1+\theta)^2$ and $\|T^{-1}\|\le(1-\theta)^{-1}$.

Proof. We shall construct by transfinite recursion, for every $\delta<\lambda$, a bounded linear operator $V_\delta:\ell_1(N_\delta)\to\ell_1(\Gamma_\delta)$ with $\|V_\delta\|\le\theta$. For $\delta\le\lambda$ and $x\in\ell_1(\Gamma_\delta)$ put

$$W_\delta(x)=x+\sum_{\delta'<\delta}V_{\delta'}(x|_{N_{\delta'}}),\qquad U_{<\delta}(x)=\sum_{\delta'<\delta}U_{\delta'}(x|_{N_{\delta'}}),\qquad T_\delta=U_{<\delta}W_\delta.$$

Since the sets $N_{\delta'}$ are pairwise disjoint and $\|x\|=\sum_{\delta'<\delta}\|x|_{N_{\delta'}}\|$, the series defining $W_\delta(x)$ converges absolutely in $\ell_1(\Gamma_\delta)$ and $\|W_\delta-1\|\le\theta$, so W_δ is a linear bijection of $\ell_1(\Gamma_\delta)$ with $\|W_\delta^{-1}\|\le(1-\theta)^{-1}$; the operator $U_{<\delta}$ is the ℓ_1-sum of the operators $U_{\delta'}$, so it is a linear bijection of $\ell_1(\Gamma_\delta)$ with $\|U_{<\delta}\|\le1+\theta$ and $\|U_{<\delta}^{-1}\|\le1$. Hence T_δ is a linear bijection of $\ell_1(\Gamma_\delta)$ with $\|T_\delta\|\le(1+\theta)^2$ and $\|T_\delta^{-1}\|\le(1-\theta)^{-1}$. Moreover, for $\delta'<\delta$ the operator T_δ coincides with $T_{\delta'}$ on $\ell_1(\Gamma_{\delta'})$, because for $x\in\ell_1(\Gamma_{\delta'})$ we have $x|_{N_{\delta''}}=0$ for $\delta''\ge\delta'$. The operators V_δ will be chosen so that

$$T_\delta(P_\delta)=\ell_1^0(\Gamma_\delta)\quad\text{for all }\delta\le\lambda.\tag{1}$$

Then $T=T_\lambda$ is the required operator.

Let $\delta<\lambda$ and suppose that $V_{\delta'}$ has been constructed for all $\delta'<\delta$ in such a way that (1) holds for all ordinals $\le\delta$ (for $\delta=0$ this is trivial, since $\Gamma_0=\varnothing$). We construct V_δ so that (1) holds for $\delta+1$; note that (1) for a limit ordinal $\mu\le\lambda$ follows from (1) for all $\delta<\mu$, because by (*ii*) and the compatibility of the operators T_δ we have $T_\mu(P_\mu)=\bigcup_{\delta<\mu}T_\delta(P_\delta)=\bigcup_{\delta<\mu}\ell_1^0(\Gamma_\delta)=\ell_1^0(\Gamma_\mu)$.

If $N_\delta = \varnothing$, then $P_{\delta+1} = P_\delta$, $T_{\delta+1} = T_\delta$, and we put $V_\delta = 0$. So let $N_\delta \neq \varnothing$, and put $U = U_\delta$ and $y_n = U^{-1}(e_n)$ for $n \in N_\delta$. By (iii), $(y_n)_{n\in N_\delta}$ is a Hamel basis of D_δ and

$$\Big\|\sum_n c_n y_n\Big\| \geq \|U\|^{-1}\Big\|\sum_n c_n e_n\Big\| \geq (1+\theta)^{-1}\sum_n |c_n|$$

for all finitely supported families $(c_n)_{n\in N_\delta}$ of scalars. For every $n \in N_\delta$ choose $x_n \in P_{\delta+1}$ with $x_n|_{N_\delta} = y_n$. Since P_δ is dense in $\ell_1(\Gamma_\delta)$ by (i), the coset $-x_n|_{\Gamma_\delta} + P_\delta$ is dense in $\ell_1(\Gamma_\delta)$, and we can choose $z_n \in \ell_1(\Gamma_\delta)$ with

$$x_n|_{\Gamma_\delta} + z_n \in P_\delta, \qquad \|z_n\| \leq \theta(1+\theta)^{-2}.$$

Define V_δ on D_δ by $V_\delta(\sum_n c_n y_n) = \sum_n c_n W_\delta(z_n)$. Then

$$\Big\|V_\delta\Big(\sum_n c_n y_n\Big)\Big\| \leq \|W_\delta\|\,\theta(1+\theta)^{-2}\sum_n |c_n| \leq \theta(1+\theta)^{-1}\sum_n |c_n| \leq \theta\Big\|\sum_n c_n y_n\Big\|,$$

so V_δ extends by continuity from the dense subspace D_δ to a bounded linear operator $V_\delta : \ell_1(N_\delta) \to \ell_1(\Gamma_\delta)$ with $\|V_\delta\| \leq \theta$ (recall that D_δ is dense in $\ell_1(N_\delta)$ by (i), since the restriction map is a contractive surjection of $\ell_1(\Gamma_{\delta+1})$ onto $\ell_1(N_\delta)$).

We claim that

$$x|_{\Gamma_\delta} + W_\delta^{-1}V_\delta(x|_{N_\delta}) \in P_\delta \quad \text{for every } x \in P_{\delta+1}. \tag{2}$$

Indeed, write $x|_{N_\delta} = \sum_n c_n y_n$ with finitely many non-zero c_n. Then $x - \sum_n c_n x_n \in P_{\delta+1}$ vanishes on N_δ, so it belongs to P_δ, and hence $x|_{\Gamma_\delta} = \sum_n c_n x_n|_{\Gamma_\delta} + p$ for some $p \in P_\delta$. Therefore

$$x|_{\Gamma_\delta} + W_\delta^{-1}V_\delta(x|_{N_\delta}) = \sum_n c_n\big(x_n|_{\Gamma_\delta} + z_n\big) + p \in P_\delta.$$

Now let $x \in P_{\delta+1}$. Since $W_{\delta+1}(x) = W_\delta(x|_{\Gamma_\delta}) + V_\delta(x|_{N_\delta}) + x|_{N_\delta}$ and $U_{<\delta+1}$ acts as $U_{<\delta}$ on $\ell_1(\Gamma_\delta)$ and as U on $\ell_1(N_\delta)$, we get

$$T_{\delta+1}(x) = U_{<\delta}W_\delta\big(x|_{\Gamma_\delta} + W_\delta^{-1}V_\delta(x|_{N_\delta})\big) + U(x|_{N_\delta}) = T_\delta\big(x|_{\Gamma_\delta} + W_\delta^{-1}V_\delta(x|_{N_\delta})\big) + U(x|_{N_\delta}).$$

By (2) and (1) for δ, the first summand belongs to $\ell_1^0(\Gamma_\delta)$, and the second one belongs to $U(D_\delta) = \ell_1^0(N_\delta)$. Hence $T_{\delta+1}(P_{\delta+1}) \subset \ell_1^0(\Gamma_{\delta+1})$. Conversely, $\ell_1^0(\Gamma_\delta) = T_\delta(P_\delta) = T_{\delta+1}(P_\delta) \subset T_{\delta+1}(P_{\delta+1})$, and for every $n \in N_\delta$ the above formula gives $T_{\delta+1}(x_n) \in \ell_1^0(\Gamma_\delta) + U(y_n) = \ell_1^0(\Gamma_\delta) + e_n$, so $e_n \in T_{\delta+1}(P_{\delta+1})$. Thus, $T_{\delta+1}(P_{\delta+1}) = \ell_1^0(\Gamma_{\delta+1})$, which completes the recursion and the proof. □

If all the sets N_δ are at most countable and $P_{\delta+1}/P_\delta$ has at most countable algebraic dimension for every $\delta < \lambda$, then condition (iii) is automatically satisfied for every $\theta > 0$: indeed, D_δ is a dense subspace of $\ell_1(N_\delta)$ of countable algebraic dimension, so for infinite N_δ theorem 4.5 provides a linear bijection U_δ with $U_\delta(D_\delta) = \ell_1^0(N_\delta)$ and $\|U_\delta\|\|U_\delta^{-1}\| \leq 1+\theta$, which can be normalized by a scalar, and for finite N_δ we have $D_\delta = \ell_1(N_\delta)$ and $U_\delta = 1_{\ell_1(N_\delta)}$ works. Example 6.3 shows that condition (ii) cannot be dropped: for the filtration $\Gamma_\delta = \delta$ of ω_1 the space P_L satisfies (i) and (iii), but fails (ii) at every limit ordinal.

We proceed to the construction of filtrations satisfying the hypotheses of theorem 6.4. For the rest of this section we fix a set Γ, a dense topologically projective subspace P of $\ell_1(\Gamma)$, a set S and bounded linear operators $\sigma : P \to \ell_1^0(S)$ and $\pi : \ell_1^0(S) \to P$ with $\pi\sigma = 1_P$ and $\|\pi\| \leq 1$; these exist by proposition 2.6. Since P is dense in $\ell_1(\Gamma)$, we can also fix for every $\gamma \in \Gamma$ a sequence $(p_n^\gamma)_{n\in\mathbb{N}}$ in P converging to e_γ. For $A \subset \Gamma$ we put $P_A = P \cap \ell_1(A)$.

Definition 6.5. *A pair (A, B) of subsets $A \subset \Gamma$ and $B \subset S$ is called closed if*

(a) $\operatorname{supp}(\pi(e_s)) \subset A$ *for all* $s \in B$;

(b) $\operatorname{supp}(\sigma(x)) \subset B$ *for all* $x \in P_A$;

(c) $\operatorname{supp}(p_n^\gamma) \subset A$ *for all* $\gamma \in A$ *and* $n \in \mathbb{N}$.

Lemma 6.6. *Let (A, B) be a closed pair. Then $P_A = \pi(\ell_1^0(B))$, the space P_A is dense in $\ell_1(A)$, and $\sigma|_{P_A} : P_A \to \ell_1^0(B)$, $\pi|_{\ell_1^0(B)} : \ell_1^0(B) \to P_A$ exhibit P_A as a retract of $\ell_1^0(B)$. In particular, P_A is topologically projective.*

Proof. By (a), $\pi(\ell_1^0(B)) \subset P \cap \ell_1(A) = P_A$. If $x \in P_A$, then $\sigma(x) \in \ell_1^0(B)$ by (b) and $x = \pi\sigma(x) \in \pi(\ell_1^0(B))$. By (c), P_A contains the sequences p_n^γ, $\gamma \in A$, so its closure contains $\ell_1^0(A)$ and P_A is dense in $\ell_1(A)$. The last claim is now obvious. □

Lemma 6.7. *Let μ be an infinite cardinal, $A \subset \Gamma$ and $B \subset S$ be sets of cardinality at most μ. Then there is a closed pair (A', B') with $A \subset A'$, $B \subset B'$ and $|A'|, |B'| \le \mu$.*

Proof. Define increasing sequences $A = A_0 \subset A_1 \subset \dots$ and $B = B_0 \subset B_1 \subset \dots$ as follows. Given A_k and B_k of cardinality at most μ, let H_k be a Hamel basis of P_{A_k}. Since $P_{A_k} \subset \ell_1(A_k)$ has density character at most μ, by theorem 3.1 $|H_k| \le \mu$. Put

$$B_{k+1} = B_k \cup \bigcup_{x \in H_k} \operatorname{supp}(\sigma(x)), \qquad A_{k+1} = A_k \cup \bigcup_{s \in B_{k+1}} \operatorname{supp}(\pi(e_s)) \cup \bigcup_{\gamma \in A_k,\, n \in \mathbb{N}} \operatorname{supp}(p_n^\gamma).$$

All supports are countable, so $|A_{k+1}|, |B_{k+1}| \le \mu$. Put $A' = \bigcup_k A_k$ and $B' = \bigcup_k B_k$. Conditions (a) and (c) hold for (A', B'), since every $s \in B'$ and every $\gamma \in A'$ belongs to some B_{k+1} or A_k respectively. To verify (b), note first that $\bigcup_k P_{A_k}$ is dense in $\ell_1(A')$: indeed, $P_{A_{k+1}}$ contains p_n^γ for all $\gamma \in A_k$, so the closure of $\bigcup_k P_{A_k}$ contains $\ell_1^0(A')$. Let $x \in P_{A'}$ and choose $x_j \in \bigcup_k P_{A_k}$ with $x_j \to x$. Then $\sigma(x_j) \in \ell_1^0(B_{k_j+1}) \subset \ell_1(B')$ for suitable k_j, because $\operatorname{supp}(\sigma(y)) \subset B_{k+1}$ for $y \in P_{A_k}$ by the choice of B_{k+1}. Since σ is continuous and $\ell_1(B')$ is closed in $\ell_1(S)$, we get $\sigma(x) \in \ell_1(B') \cap \ell_1^0(S) = \ell_1^0(B')$. □

Note that in the last argument the support of $x \in P_{A'}$ need not be contained in a single A_k; this is why density was used instead of a direct argument. The same argument proves the following lemma.

Lemma 6.8. *Let $((A_\delta, B_\delta))_{\delta<\lambda}$ be an increasing family of closed pairs indexed by a limit ordinal λ, and let $A_\lambda = \bigcup_{\delta<\lambda} A_\delta$, $B_\lambda = \bigcup_{\delta<\lambda} B_\delta$. Then (A_λ, B_λ) is a closed pair and $P_{A_\lambda} = \bigcup_{\delta<\lambda} P_{A_\delta}$.*

Proof. Conditions (a) and (c) are inherited from the pairs (A_δ, B_δ). By lemma 6.6 each P_{A_δ} is dense in $\ell_1(A_\delta)$, so $\bigcup_{\delta<\lambda} P_{A_\delta}$ is dense in $\ell_1(A_\lambda)$. If $x \in P_{A_\lambda}$, choose $x_j \in \bigcup_\delta P_{A_\delta}$ with $x_j \to x$; then $\sigma(x_j) \in \ell_1(B_\lambda)$ by (b) for the pairs (A_δ, B_δ), hence $\sigma(x) \in \ell_1(B_\lambda) \cap \ell_1^0(S) = \ell_1^0(B_\lambda)$. This is (b) for (A_λ, B_λ). Finally, by lemma 6.6, $P_{A_\lambda} = \pi(\ell_1^0(B_\lambda)) = \bigcup_{\delta<\lambda} \pi(\ell_1^0(B_\delta)) = \bigcup_{\delta<\lambda} P_{A_\delta}$. □

Lemma 6.9. *Let $(A, B) \subset (A', B')$ be closed pairs, that is, $A \subset A'$ and $B \subset B'$, and let $N = A' \setminus A$. Then $D = \{x|_N : x \in P_{A'}\}$ is a dense subspace of $\ell_1(N)$ which is a retract of $\ell_1^0(B' \setminus B)$. In particular, D is topologically projective.*

Proof. The restriction map $r : \ell_1(A') \to \ell_1(N)$, $x \mapsto x|_N$, is a contractive surjection and $P_{A'}$ is dense in $\ell_1(A')$ by lemma 6.6, so $D = r(P_{A'})$ is dense in $\ell_1(N)$. The kernel of $r|_{P_{A'}}$ is $P_{A'} \cap \ell_1(A) = P_A$.

Let $y \in D$ and $x \in P_{A'}$ with $x|_N = y$. By (b) for (A', B') we have $\sigma(x) \in \ell_1^0(B')$. If $x' \in P_{A'}$ is another element with $x'|_N = y$, then $x - x' \in P_A$, so $\sigma(x - x') \in \ell_1^0(B)$ by (b) for (A, B), and $\sigma(x)|_{S\setminus B} = \sigma(x')|_{S\setminus B}$. Therefore the map

$$\rho : D \to \ell_1^0(B' \setminus B), \qquad \rho(y) = \sigma(x)|_{S\setminus B} \quad (x \in P_{A'},\ x|_N = y),$$

is well defined; it is linear, because if $x_1, x_2 \in P_{A'}$ restrict to y_1, y_2, then $x_1 + cx_2$ restricts to $y_1 + cy_2$. It is bounded: given $y \in D$, $x_0 \in P_{A'}$ with $x_0|_N = y$ and $\varepsilon > 0$, use the density of P_A in $\ell_1(A)$ to find $p \in P_A$ with $\|x_0|_A + p\| \le \varepsilon$; then $x = x_0 + p \in P_{A'}$ satisfies $x|_N = y$ and $\|x\| = \|x|_A\| + \|y\| \le \|y\| + \varepsilon$, so $\|\rho(y)\| \le \|\sigma(x)\| \le \|\sigma\|(\|y\| + \varepsilon)$. Hence $\|\rho\| \le \|\sigma\|$. Finally, let $r\pi : \ell_1^0(B' \setminus B) \to \ell_1(N)$ be the composition of π with the restriction to N. For $y = x|_N$ as above,

$$r\pi\rho(y) = r\big(\pi\sigma(x) - \pi(\sigma(x)|_B)\big) = r(x) - r\big(\pi(\sigma(x)|_B)\big) = y,$$

because $\pi(\sigma(x)|_B) \in \pi(\ell_1^0(B)) = P_A \subset \ell_1(A)$ vanishes on N. Thus, $r\pi$ maps $\ell_1^0(B' \setminus B)$ into D (its range is contained in $r(P_{A'}) = D$) and $r\pi\rho = 1_D$; so D is a retract of $\ell_1^0(B' \setminus B)$ and is topologically projective by proposition 2.5. □

Theorem 6.10. *Let Γ be a set and P be a dense topologically projective subspace of $\ell_1(\Gamma)$. Then for every $\theta > 0$ there is a linear bijection T of $\ell_1(\Gamma)$ with $T(P) = \ell_1^0(\Gamma)$, $\|T\| \le 1 + \theta$ and $\|T^{-1}\| \le 1$. In particular, $d_{BM}(P, \ell_1^0(\Gamma)) = 1$ and P is extremally projective.*

Proof. We proceed by induction on the cardinality κ of Γ. If Γ is finite, then $P = \ell_1(\Gamma) = \ell_1^0(\Gamma)$ and $T = 1_{\ell_1(\Gamma)}$ works. If Γ is countably infinite, then P has countable algebraic dimension by theorem 3.1, and theorem 4.5 provides a linear bijection T with $T(P) = \ell_1^0(\Gamma)$ and $\|T\|\|T^{-1}\| \le 1 + \theta$; multiplying T by a scalar we get $\|T^{-1}\| \le 1$ and $\|T\| \le 1 + \theta$.

Let κ be uncountable and assume that the theorem holds for all sets of cardinality less than κ. Since the conclusion for a smaller θ is stronger, we may fix $\theta \in (0, 1/2)$; put $\theta' = \theta/8$, so that $(1 + \theta')^2(1 - \theta')^{-1} \le 1 + \theta$. Fix S, σ, π and the sequences p_n^γ as above. Let λ be the cofinality of κ and $(\Gamma^\delta)_{\delta<\lambda}$ be a family of subsets of Γ of cardinality less than κ with $\bigcup_{\delta<\lambda} \Gamma^\delta = \Gamma$. Such a family exists: if κ is regular, then $\lambda = \kappa$ and we may take $\Gamma^\delta = \{\gamma_\delta\}$ for an enumeration $(\gamma_\delta)_{\delta<\kappa}$ of Γ; if κ is singular, we take an enumeration $(\gamma_\alpha)_{\alpha<\kappa}$ of Γ, an increasing sequence of cardinals $(\kappa_\delta)_{\delta<\lambda}$ with supremum κ, and $\Gamma^\delta = \{\gamma_\alpha : \alpha < \kappa_\delta\}$. By transfinite recursion we define an increasing family $((A_\delta, B_\delta))_{\delta\le\lambda}$ of closed pairs with $|A_\delta|, |B_\delta| < \kappa$ for $\delta < \lambda$: put $A_0 = B_0 = \varnothing$; given (A_δ, B_δ) with $\delta < \lambda$, let $(A_{\delta+1}, B_{\delta+1})$ be a closed pair containing $(A_\delta \cup \Gamma^\delta, B_\delta)$ with $|A_{\delta+1}|, |B_{\delta+1}| \le \max(|A_\delta|, |B_\delta|, |\Gamma^\delta|, \aleph_0) < \kappa$, which exists by lemma 6.7; for a limit ordinal $\delta \le \lambda$ let A_δ and B_δ be the unions of the previous sets. For a limit ordinal $\delta < \lambda$ the set A_δ is a union of fewer than $\mathrm{cf}(\kappa)$ sets of cardinality less than κ, so $|A_\delta| < \kappa$, and similarly $|B_\delta| < \kappa$; by lemma 6.8 the pairs (A_δ, B_δ) are closed for all $\delta \le \lambda$. Finally, $A_\lambda = \Gamma$.

Thus, $(A_\delta)_{\delta\le\lambda}$ is a filtration of Γ, and we verify the hypotheses of theorem 6.4 for this filtration and θ', where $P_\delta = P_{A_\delta}$ and $N_\delta = A_{\delta+1} \setminus A_\delta$. Condition (i) holds by lemma 6.6, and condition (ii) holds by lemma 6.8. For (iii), let $\delta < \lambda$. By lemma 6.9 the space $D_\delta = \{x|_{N_\delta} : x \in P_{\delta+1}\}$ is a dense topologically projective subspace of $\ell_1(N_\delta)$, and $|N_\delta| \le |A_{\delta+1}| < \kappa$. By the induction hypothesis there is a linear bijection U_δ of $\ell_1(N_\delta)$ with $U_\delta(D_\delta) = \ell_1^0(N_\delta)$, $\|U_\delta\| \le 1 + \theta'$ and $\|U_\delta^{-1}\| \le 1$ (if $N_\delta = \varnothing$ there is nothing to do). Now theorem 6.4 gives a linear bijection T of $\ell_1(\Gamma)$ with $T(P) = \ell_1^0(\Gamma)$ and $\|T\|\|T^{-1}\| \le (1 + \theta')^2(1 - \theta')^{-1} \le 1 + \theta$; after multiplication by a scalar, $\|T^{-1}\| \le 1$ and $\|T\| \le 1 + \theta$. The last claim of the theorem follows from theorem 2.7. □

Note that the filtration constructed in the proof of theorem 6.10 has blocks N_δ of cardinality less than $|\Gamma|$, not necessarily countable ones; this is why theorem 6.4 was formulated for arbitrary blocks.

Together with theorem 4.1, theorem 6.10 gives the main result of the paper.

Theorem 6.11. *For a normed space P the following conditions are equivalent:*

(i) P is extremally projective;

(ii) P is topologically projective and $\overline{P}$ is isometrically isomorphic to $\ell_1(\Gamma)$ for some set Γ;

(iii) $d_{BM}(P, \ell_1^0(\Gamma)) = 1$ for some set Γ;

(iv) for every $\varepsilon > 0$ there is a set Λ with $d_{BM}(P, \ell_1^0(\Lambda)) < 1 + \varepsilon$.

Moreover, in (iii) the set Γ necessarily has the cardinality of the algebraic dimension of P if P is infinite dimensional, and the Banach–Mazur distance in (iii) is realized by linear bijections $T : \overline{P} \to \ell_1(\Gamma)$ with $T(P) = \ell_1^0(\Gamma)$ and $\|T\|\|T^{-1}\|$ arbitrarily close to 1.

Proof. $(i) \implies (ii)$ Extremally projective spaces are topologically projective, and their completions are described by theorem 4.1.

$(ii) \implies (iii)$ Let $J : \overline{P} \to \ell_1(\Gamma)$ be an isometric isomorphism. Then $J(P)$ is a dense topologically projective subspace of $\ell_1(\Gamma)$, and theorem 6.10 gives $d_{BM}(P, \ell_1^0(\Gamma)) = d_{BM}(J(P), \ell_1^0(\Gamma)) = 1$, realized by linear bijections of the required form.

$(iii) \implies (iv)$ is obvious, and $(iv) \implies (i)$ follows from theorem 2.7, since a linear bijection $S : P \to \ell_1^0(\Lambda)$ with $\|S\|\|S^{-1}\| < 1 + \varepsilon$ makes P a $(1+\varepsilon)$-retract of $\ell_1^0(\Lambda)$. Finally, if P is infinite dimensional and $d_{BM}(P, \ell_1^0(\Gamma)) < \infty$, then P and $\ell_1^0(\Gamma)$ are isomorphic as vector spaces, and the algebraic dimension of $\ell_1^0(\Gamma)$ is the cardinality of Γ. □

Corollary 6.12. *Every extremally projective normed space P is almost isometric to $\ell_1^0(\Gamma)$, where Γ is a set whose cardinality is the algebraic dimension of P. In particular, the conjecture of Grønbæk [9, conjecture 3.1] holds in the almost isometric form, while corollary 5.3 shows that it fails in the isometric form.*

We finish with some comments. Let us compare our argument with the proof of the theorem of Grønbæk, which consists of two steps: a transfinite construction which finds in every subspace Y of $\ell_1^0(\Lambda)$ a complemented subspace isomorphic to $\ell_1^0(\Lambda')$, where the cardinality of Λ' is the density character of Y [9, proposition 2.2], and the decomposition method of Pełczyński [9, proposition 2.3]. An inspection of the first step shows that it produces, for every $\varepsilon > 0$, a subspace which is $(1+\varepsilon)$-isomorphic to $\ell_1^0(\Lambda')$ and $(1+\varepsilon)$-complemented, so the constants are lost only in the second step. Theorems 4.5 and 6.4 replace the decomposition method by back-and-forth constructions which keep the constants, at the price of the continuity condition (ii), and lemmas 6.7 and 6.8 show that this condition can always be achieved for topologically projective spaces. Next, theorem 6.11 shows that for dense subspaces of the spaces $\ell_1(\Gamma)$ topological and extremal projectivity coincide. This is a feature of the exact norm of $\ell_1(\Gamma)$: if E is an infinite dimensional Banach space which is topologically but not almost isometrically isomorphic to $\ell_1(\Gamma)$, for instance $\ell_1(\Gamma)$ with a suitable equivalent norm, then every dense subspace of E of the form $J(\ell_1^0(\Gamma))$, J a topological isomorphism, is topologically but not extremally projective by theorem 4.1. Further, by theorem 6.11 and example 6.3, the question of Helemskii on dense subspaces of $\ell_1(\Lambda)$ has the following answer: a dense subspace P of $\ell_1(\Lambda)$ is extremally projective if and only if it is topologically projective, if

and only if it admits a filtration with the properties (i)–(iii) of theorem 6.4; for countable Λ this means that P has countable algebraic dimension, but for uncountable Λ the continuity condition (ii) is a genuine restriction, as the ladder space shows. Finally, Fodor's lemma is used in this paper only in the negative direction, to show that the ladder space is not topologically projective; the positive results use only the closing-off arguments of lemmas 6.7 and 6.8. We do not know an intrinsic description of topologically projective dense subspaces of $\ell_1(\Lambda)$ for uncountable Λ which would not refer to a retraction onto a free space.

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